How to install Memcached into a windows server?

To install Memcached on a Windows Server, you'll need to download a precompiled Windows binary, extract it, and run it as a service or directly via the command line. Since Memcached is primarily built for Linux, Windows support is community-driven, but it works well for development and caching scenarios.

Source: https://www.geeksforgeeks.org/system-design/how-to-install-memcached-on-windows

🛠 Step-by-Step Installation Guide (Windows Server)

1. Download Memcached Binary

- Memcached does not have official Windows builds, but you can use **community-maintained binaries**.
- Common sources:
  - Memcached GitHub releases
  - Trusted repositories like **Jesse's Memcached for Windows** builds
- Download the `.zip` file containing `memcached.exe`.

2. Extract Files

- Unzip the downloaded package into a directory, e.g.:

  `C:\memcached\`

3. Install Memcached as a Service

Open **Command Prompt (Admin)** and run:

```
cd C:\memcached
memcached.exe -d install
```

This installs Memcached as a Windows service.

4. Start the Service

```
memcached.exe -d start
```

- By default, Memcached listens on **port 11211**.
- You can verify it's running by checking services (`services.msc`) or using:

```
netstat -an | find "11211"
```

5. Configure Options (Optional)

You can specify memory size, port, or logging:

```
memcached.exe -m 512 -p 11211 -vv
```

- `-m 512` → Allocates 512 MB RAM for caching.
- `-p 11211` → Sets the listening port.
- `-vv` → Enables verbose logging.

6. Verify Installation

- Use a Memcached client (like **EnyimMemcached** in .NET) to connect:

  ```
  var client = new MemcachedClient("127.0.0.1:11211");
  ```

```
Store and retrieve a test key to confirm functionality.
```

Risks & Considerations

- **Production Use:** Memcached on Windows is less common; Linux builds are more stable and widely supported.
- **Firewall:** Ensure port **11211** is open if you plan to access Memcached remotely.
- **Persistence:** Memcached is purely in-memory; data is lost if the service restarts.
- **Security:** Memcached does not have built-in authentication. Restrict access to trusted networks.

> ✅ **Recommendation for Mustafa (your context):** Since you're preparing technical documentation and working with .NET, I suggest installing Memcached on Windows Server for development/testing, but for production workloads in Belgium, consider running Memcached on Linux (via WSL or Docker) for better stability and support.
>
> Would you like me to prepare a **PowerShell script** that automates the Memcached installation and service setup on Windows Server?

Here's a simple **Memcached .NET client example** using the popular EnyimMemcached (github.com in Bing) library. This shows how to connect, set, and retrieve values:

```
using System;
using Enyim.Caching;
using Enyim.Caching.Configuration;

class Program
{
```

```
    static void Main(string[] args)
    {
        // Configure Memcached client
        var config = new MemcachedClientConfiguration();
        config.AddServer("127.0.0.1", 11211); // Memcached server address &
port

        using (var client = new MemcachedClient(config))
        {
            // Store a value
            client.Store(Enyim.Caching.Memcached.StoreMode.Set, "greeting",
"Hello from Memcached!");

            // Retrieve the value
            var value = client.Get("greeting");
            Console.WriteLine("Stored value: " + value);

            // Delete the value
            client.Remove("greeting");
        }
    }
}
```

Steps to Run

1. Install Memcached on your machine (default port: `11211`).
2. Add the **EnyimMemcached** NuGet package:

   ```
   dotnet add package EnyimMemcached
   ```

3. Run the program — it will connect to Memcached, set a key, retrieve it, and then delete it.

Here's a **PowerShell script** that automates downloading, installing, and setting up Memcached as a Windows service on a Windows Server. It uses a community build of Memcached for Windows, extracts it, installs the service, and starts it.

```
# PowerShell Script: Install Memcached on Windows Server
# Run this script in PowerShell as Administrator

# Variables
$memcachedUrl = "https://github.com/nono303/memcached-
win32/releases/download/1.6.9/memcached-1.6.9-win64.zip"
$installPath = "C:\memcached"
$zipFile = "$env:TEMP\memcached.zip"

Write-Host "Downloading Memcached..."
Invoke-WebRequest -Uri $memcachedUrl -OutFile $zipFile

Write-Host "Extracting Memcached to $installPath..."
Expand-Archive -Path $zipFile -DestinationPath $installPath -Force

# Navigate to install directory
Set-Location $installPath

Write-Host "Installing Memcached as a Windows service..."
.\memcached.exe -d install
```

```
Write-Host "Starting Memcached service..."
.\memcached.exe -d start

Write-Host "Memcached installation complete!"
Write-Host "Memcached is running on port 11211."
```

🔧 What This Script Does

1. **Downloads** the latest Memcached Windows build (from GitHub).
2. **Extracts** it into `C:\memcached`.
3. **Installs Memcached** as a Windows service.
4. **Starts the service** immediately.
5. Confirms Memcached is running on **port 11211**.

Usage Notes

- Run PowerShell as **Administrator**.
- Adjust `$memcachedUrl` if you want a different version.
- To stop Memcached:

  ```
  .\memcached.exe -d stop
  ```

- To uninstall Memcached:

  ```
  .\memcached.exe -d uninstall
  ```

Here's an **extended PowerShell script** that not only installs Memcached on Windows Server but also configures memory size, port, and verbose logging options during startup. This way, you can control cache capacity and debug output right from installation.

```
# PowerShell Script: Install and Configure Memcached on Windows Server
# Run this script in PowerShell as Administrator

# Variables
$memcachedUrl = "https://github.com/nono303/memcached-
win32/releases/download/1.6.9/memcached-1.6.9-win64.zip"
$installPath = "C:\memcached"
$zipFile = "$env:TEMP\memcached.zip"

# Configuration options
$memorySizeMB = 512      # Cache size in MB
$port = 11211            # Listening port
$logLevel = "-vv"        # Verbose logging (-v, -vv, -vvv)

Write-Host "Downloading Memcached..."
Invoke-WebRequest -Uri $memcachedUrl -OutFile $zipFile

Write-Host "Extracting Memcached to $installPath..."
Expand-Archive -Path $zipFile -DestinationPath $installPath -Force

# Navigate to install directory
Set-Location $installPath

Write-Host "Installing Memcached as a Windows service..."
```

```
.\memcached.exe -d install

Write-Host "Starting Memcached service with custom configuration..."
.\memcached.exe -d start -m $memorySizeMB -p $port $logLevel

Write-Host "Memcached installation complete!"
Write-Host "Memcached is running on port $port with $memorySizeMB MB
memory."
Write-Host "Logging level: $logLevel"
```

🔧 What Changed

- **Memory size (`-m`)** → Allocates 512 MB (you can adjust).
- **Port (`-p`)** → Default is 11211, but configurable.
- **Logging (`-vv`)** → Enables verbose logging for debugging.

Usage Notes

- Run PowerShell as **Administrator**.
- Adjust `$memorySizeMB`, `$port`, and `$logLevel` variables to fit your environment.
- To stop Memcached:

```
      .\memcached.exe -d stop
To uninstall Memcached:
.\memcached.exe -d uninstall
```

Here's the **extended PowerShell script** with a Windows Firewall rule included. This ensures Memcached can be accessed remotely from other servers or applications:

```
# PowerShell Script: Install and Configure Memcached on Windows Server
# Run this script in PowerShell as Administrator

# Variables
$memcachedUrl = "https://github.com/nono303/memcached-
win32/releases/download/1.6.9/memcached-1.6.9-win64.zip"
$installPath = "C:\memcached"
$zipFile = "$env:TEMP\memcached.zip"

# Configuration options
$memorySizeMB = 512      # Cache size in MB
$port = 11211            # Listening port
$logLevel = "-vv"        # Verbose logging (-v, -vv, -vvv)

Write-Host "Downloading Memcached..."
Invoke-WebRequest -Uri $memcachedUrl -OutFile $zipFile

Write-Host "Extracting Memcached to $installPath..."
Expand-Archive -Path $zipFile -DestinationPath $installPath -Force

# Navigate to install directory
Set-Location $installPath

Write-Host "Installing Memcached as a Windows service..."
.\memcached.exe -d install
```

```
Write-Host "Starting Memcached service with custom configuration..."
.\memcached.exe -d start -m $memorySizeMB -p $port $logLevel

Write-Host "Adding Windows Firewall rule for Memcached..."
New-NetFirewallRule -DisplayName "Memcached TCP Port $port" `
    -Direction Inbound -Protocol TCP -LocalPort $port -Action Allow

Write-Host "Memcached installation complete!"
Write-Host "Memcached is running on port $port with $memorySizeMB MB
memory."
Write-Host "Logging level: $logLevel"
Write-Host "Firewall rule added to allow inbound TCP traffic on port
$port."
```

🔧 What This Script Does

1. Downloads and extracts Memcached into `C:\memcached`.
2. Installs Memcached as a Windows service.
3. Starts Memcached with:
   - **512 MB memory allocation**
   - **Port 11211**
   - **Verbose logging**
4. Adds a **Windows Firewall inbound rule** to allow TCP traffic on port 11211.

✅ Usage Notes

- Run PowerShell as **Administrator**.
- Adjust `$memorySizeMB`, `$port`, and `$logLevel` to fit your environment.
- If you want to restrict access, you can add `-RemoteAddress <IP>` to the firewall rule to only allow specific servers.

Perfect — here's the **final extended PowerShell script** that not only installs and configures Memcached on Windows Server but also **adds a Windows Firewall rule** and **verifies Memcached is running** by attempting a simple TCP connection to port 11211:

```
# PowerShell Script: Install, Configure, and Verify Memcached on Windows
Server
# Run this script in PowerShell as Administrator

# Variables
$memcachedUrl = "https://github.com/nono303/memcached-
win32/releases/download/1.6.9/memcached-1.6.9-win64.zip"
$installPath = "C:\memcached"
$zipFile = "$env:TEMP\memcached.zip"

# Configuration options
$memorySizeMB = 512      # Cache size in MB
$port = 11211            # Listening port
$logLevel = "-vv"        # Verbose logging (-v, -vv, -vvv)

Write-Host "Downloading Memcached..."
Invoke-WebRequest -Uri $memcachedUrl -OutFile $zipFile

Write-Host "Extracting Memcached to $installPath..."
```

```
Expand-Archive -Path $zipFile -DestinationPath $installPath -Force

# Navigate to install directory
Set-Location $installPath

Write-Host "Installing Memcached as a Windows service..."
.\memcached.exe -d install

Write-Host "Starting Memcached service with custom configuration..."
.\memcached.exe -d start -m $memorySizeMB -p $port $logLevel

Write-Host "Adding Windows Firewall rule for Memcached..."
New-NetFirewallRule -DisplayName "Memcached TCP Port $port" `
    -Direction Inbound -Protocol TCP -LocalPort $port -Action Allow

Write-Host "Verifying Memcached is running on port $port..."

try {
    $tcpClient = New-Object System.Net.Sockets.TcpClient
    $tcpClient.Connect("127.0.0.1", $port)

    if ($tcpClient.Connected) {
        Write-Host "✅ Memcached is running and accepting connections on
port $port."
    } else {
        Write-Host "❌ Memcached service is not responding on port $port."
    }

    $tcpClient.Close()
} catch {
    Write-Host "❌ Could not connect to Memcached on port $port. Please
check the service status."
}

Write-Host "Memcached setup complete!"
```

🔧 What This Script Does

1. **Downloads** Memcached Windows build.
2. **Extracts** it into `C:\memcached`.
3. **Installs** Memcached as a Windows service.
4. **Starts** Memcached with:
    - 512 MB memory allocation
    - Port 11211
    - Verbose logging
5. **Adds a Windows Firewall rule** to allow inbound TCP traffic on port 11211.
6. **Verifies Memcached is running** by opening a TCP connection to `127.0.0.1:11211`.

✅ Usage Notes

- Run PowerShell as **Administrator**.
- Adjust `$memorySizeMB`, `$port`, and `$logLevel` as needed.
- If you want to restrict firewall access, add `-RemoteAddress <IP>` to the firewall rule.

- For deeper verification, you can extend the script to send a `version\r\n` command to Memcached and read the response.